Nafiz Imtiaz Khan

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Summary

PhD candidate in Computer Science at UC Davis, advised by Professor Vladimir Filkov. Research focuses on open-source software sustainability, health informatics, and applied data science. Specializes in fine-tuning deep neural networks and large language models, building agentic retrieval-augmented generation (RAG) systems, and developing task-specific AI/ML pipelines.

Education

University of California, Davis September 2023 – Present
PhD in Computer Science | MSc in Computer Science (2025) *California, United States*
Master's degree conferred July 2025, CGPA: 3.91/4.00

Military Institute of Science and Technology (MIST)
B.Sc. in Computer Science and Engineering *Dhaka, Bangladesh*

Technical Skills

Programming Languages: Python, SQL, C/C++, Java, JavaScript, PL/SQL, Typescript, HTML / CSS, Bash
Frameworks/Libraries : PyTorch, Keras, Tensorflow, Hugging Face Transformers, LangChain, Scikit-learn, Pandas, NumPy, Matplotlib, BeautifulSoup, Pyvis, Node, React
ML/AI Techniques: Transformer architectures, fine-tuning, PEFT, LoRA/QLoRA, multimodal & vision-language models (VLMs), retrieval-augmented generation (RAG), benchmarking & evaluation
Tools/Platform: Ollama, Flask, Git, AWS Bedrock, Google Colab, GCP, Docker, Kubernetes, Grafana, Selenium, Tableau, Microsoft Azure, JIRA
Database: ChromaDB, FAISS, MongoDB, FireBase, MySQL, Oracle, SQLite, Neo4j

Professional Experience

Artificial Intelligence Intern June 2026 – Present
BetterHelp *San Jose, California, United States (Remote)*

- Contributing to AI/ML initiatives as part of the Artificial Intelligence Intern team at BetterHelp.

Human Frontier Collective Specialist – GenAI (Internship) August 2025 – September 2025
Scale AI, Inc. *Remote – California, United States*

- Designed and critiqued complex, domain-specific problem sets to rigorously test cutting-edge generative AI models, identifying limitations and refining performance.
- Collaborated with Scale researchers in interdisciplinary sessions to analyze model behavior, shape AI research directions, and explore experimental applications.

Graduate Student Researcher September 2023 – Present
University of California, Davis *California, United States*

- Built **AngioVision** (https://github.com/Nafiz43/AngioVision), a multimodal **large language model (LLM)** pipeline for angiography report generation and validation; fine-tuned a **vision-language** framework integrating a **CLIP** alignment module, embedding, and decoder layers, achieving a **10% F1 gain** over the prior SOTA, **Claude 4.6**.
- Led the end-to-end development of **RepoWise** (https://repowise.github.io/RepoWise-website/), a domain-aware **retrieval-augmented generation (RAG)** system for OSS governance analytics; benchmarked and demonstrated 84% accuracy on factual queries, outperforming general-purpose LLMs (**GPT-4o**, **Claude 4.5 Sonnet**) on both factual and non-factual repository-analysis tasks.
- Spearheaded the development of **OSSPREY** (https://oss-prey.github.io/OSSPREY-Website/), a tool to assess open-source project health via real-time analytics with an average `latency` of 3.5s and 124MB `memory usage` per project, integrating **Vue.js** (frontend), **TensorFlow** (deep learning), and **LangChain** & **Ollama** (LLM inference pipeline design)
- Engineered **ReACT-GPT** (https://github.com/Nafiz43/ReACT-GPT), an **LLM**-and-**RAG**-powered framework for extracting actionable insights from scientific articles, achieving a 75% `BERTScore` by integrating **LangChain** (orchestration), **ChromaDB** (vector retrieval), and **Ollama** (lightweight LLM inference)
- Built **EvidenceBot** (https://nafiz43.github.io/EvidenceBot/), a privacy-preserving **RAG** tool for querying and extracting insights from a diverse set of documents, using **Streamlit** and the **ReACT-GPT** framework
- Created **PCL-Fetcher** (https://nafiz43.github.io/PCL-Fetcher/), an LLM-powered pipeline automating procedural case log documentation from radiology reports, achieving an 86% `F1 score`—exceeding the crosswalk benchmark by over 13%
- Designed **MammoGen-RAG** (https://github.com/Nafiz43/MammoGen-RAG), a multimodal RAG pipeline leveraging **vision-language models (VLMs)** to automate mammography report generation, achieving a 90% `F1 score` through integrated visual and textual reasoning
- Improved foundational language model performance in **LLM-as-Developers** (https://github.com/zeerakb1/llm-developers-evaluation) by 10% (`BERTScore`) on answering StackOverflow questions via **parameter-efficient fine-tuning (PEFT)** and **QLoRA**
- Developed and optimized **Transformer**- and **LSTM**-based models for forecasting Open Source Software (OSS) project sustainability (https://github.com/OSS-PREY/OSSPREY-Pex-Forecaster), achieving an 8% improvement in `F1 score` over SOTA

Software Engineer July 2022 – August 2023
Center for Advanced Computing Research (CACR) *Dhaka, Bangladesh*

- Built the **AFMC Admission Test** Module, a full-stack software solution for the AFMC Entrance exam, utilizing **Node.js**, **Bootstrap**, and **MongoDB** to generate unique randomized questions for 30,000 candidates and automate the evaluation and ranking process
- Spearheaded a team to develop **C-Archive** using **PHP**, **MySQL**, **SQL**, and **Bootstrap** to store ongoing events of the department [GITHUB]

Patents

1. **Nafiz Imtiaz Khan**, Vladimir Filkov, "Systems And Methods For Artificial Intelligence-Driven Evidence-Based Action" (USPTO Filing ID: 32417/70825P) [REF]

Award & Honors

- **Best Paper Award** for "RepoWise: A Conversational Framework for Open Source Repository Analysis with Retrieval-Augmented Generation and Structured Analytics," by Sankalp Kashyap, Arjun Ashok, **Nafiz Imtiaz Khan**, and Vladimir Filkov, at the 2nd International Conference on Software Engineering of Emerging Technologies **(SEET 2026)**
- **Champion** in the **Medical Robotics Challenge for Contagious Diseases**, organized by **Imperial College London** [REF]
- **Champion** in the Creative App Contest of the **Tri Robo Cup Championship** organized by the **MIST Robotics club** [REF]
- **3rd runner-up** at **COVID-19 Combatants Unification Competition**, organized by **IEEE Bangladesh Chapter** [REF]
- **Top Downloaded Article Award** - in the year 2021, and 2022; provided by **Engineering Reports Journal - Wiley**
- GGCS Summer Ph.D **Fellowship** - 2024, awarded by Graduate Group of Computer Science, UC Davis
- Recognized for academic excellence at MIST, achieving placement on the **Dean's List of Honor** and receiving the **Merit Scholarship** in 2018, 2019, and 2020

Publications [Google Scholar]

1. **Nafiz Imtiaz Khan**, Priyal Soni, Arjun Ashok, Vladimir Filkov, "OSSPREY: AI-Driven Forecasting and Intervention for OSS Project Sustainability," ASE 2025 Tool Demonstrations (IEEE/ACM International Conference on Automated Software Engineering), 2025. [LINK]
2. **Nafiz Imtiaz Khan**, Kiley Cleland, Vladimir Filkov, and Roger Eric Goldman, "Intelligent Documentation in Medical Education: Can AI Replace Manual Case Logging?," JAMIA Open, 2026. [LINK]
3. Arjun Ashok*, **Nafiz Imtiaz Khan***, Swati Singhvi, Stefan Stanciulescu, Zhouhao Wang, Vladimir Filkov (*equal contribution), "Multi-Ecosystem Modeling of OSS Project Sustainability," Journal of Systems and Software (JSS), 2026. [LINK]
4. **Nafiz Imtiaz Khan**, Vladimir Filkov, "EvidenceBot: A Privacy-Preserving, Customizable RAG-Based Tool for Enhancing Large Language Model Interactions," FSE 2025 Tool Demonstrations (Foundations of Software Engineering), 2025. [LINK]
5. **Nafiz Imtiaz Khan**, Vladimir Filkov, Roger Eric Goldman, "AI-Driven Automation of Procedural Case Log Documentation," AMIA 2025 Clinical Informatics Conference (abstract), 2025. [LINK]
6. **Nafiz Imtiaz Khan**, Vladimir Filkov, "From Models to Practice: Enhancing OSS Project Sustainability with Evidence-Based Advice," FSE 2024 (Foundations of Software Engineering), 2024. [LINK]
7. **Nafiz Imtiaz Khan**, Vladimir Filkov, "Leveraging Language Models to Discover Evidence-Based Actions for OSS Sustainability," (Under review). [LINK]
8. Sankalp Kashyap, Arjun Ashok, **Nafiz Imtiaz Khan**, Vladimir Filkov, "RepoWise: A Conversational Framework for Open Source Repository Analysis with Retrieval-Augmented Generation and Structured Analytics," Software Engineering of Emerging Technologies, Communications in Computer and Information Science (CCIS), Springer, pp. 268-288, 2026. **(Best Paper Award)** [LINK]
9. **Nafiz Imtiaz Khan**, Tahasin Mahmud, and Muhammad Nazrul Islam, "COVID-19 and Black Fungus: Analysis of Public Perceptions through Machine Learning," Engineering Reports, 2022. **(IF = 2 and SJR Rank = Q2)** [LINK]
10. Zeerak Babar, **Nafiz Imtiaz Khan**, Muhammad Hassanain, Vladimir Filkov, "Open-Source LLMs for Technical Q&A: Lessons from StackExchange," International Conference on Software Engineering of Emerging Technologies (SEET), Springer, pp. 615-626, 2025. [LINK]
11. Lin Zhou, **Nafiz Imtiaz Khan**, Charlie Schweik, Vladimir Filkov, "Open-Source Software Sustainability: Defining and Measuring a Multifaceted Concept," (Under review). [LINK]
12. Amrin Rahman, Mahedi Kamal, Afrina Zinia, Amrin Akhter Pinky, Sabrina Afrin Toma, Abedur Rahman Khan, M Akhtaruzzaman, **Nafiz Imtiaz Khan**, Muhammad Nazrul Islam, "AI Integrated Entertaining and Monitoring System for Daily Care of Elders," Proceedings of the 3rd International Conference on Computing Advancements, 2024. [LINK]